

Udit Ramawat

Senior Technical Program Leader | AI Products, Aviation Software, Frontier Hardware
ramawat.fyi

Summary

Senior technical program leader specializing in AI-native products, aviation software, spatial computing, and frontier hardware/software systems. Leads cross-functional execution from ambiguous research, architecture, and operational needs through integrated plans, requirements, release readiness, and delivery. Proven record aligning 60+ engineering teams and 120+ stakeholders across AI/ML, application software, and hardware lifecycles. Hands-on builder who uses AI coding agents to prototype internal tools and workflow automations that accelerate engineering execution.

Experience

Archer Aviation

Sr. Staff Program Manager, AI Products

San Jose, CA

July 2025 - Present

- Lead cross-functional execution for Archer Intelligence across AI research, application software, and hardware; streamline model-evaluation and MLOps workflows with Research to turn progress into operational capabilities.
- Established a weekly engineering release discipline from ad hoc releases, introducing defined QA gates, dependency reviews, and clearer planning that improve developer experience and execution predictability.
- Enabled Trivy and Dependabot scanning across rapidly evolving repositories; built a Claude skill that guides engineers through every open alert in one triage-and-cleanup sweep, reducing backend validation time from about one day to 20 minutes.
- Drive end-to-end hardware program plans from prototype through xVT validation, including requirements alignment, BOM management, integrated schedules, test planning, and lifecycle readiness.
- Built and shipped an AI-assisted internal web application that enabled flight-test engineers to define dynamic alerts at scale; reduced telemetry-analysis time-to-insight by 98%+, from ~36 hours to under 15 minutes.

Apple

Staff Engineering Program Manager, Vision Products Group

Cupertino, CA

Oct 2024 - July 2025

- Led product definition for real-time platform software architecture on next-generation spatial computing systems, spanning machine perception, sensor-driven experiences, and on-device intelligence.
- Aligned roadmap dependencies across 60+ engineering teams to de-risk delivery of launch-critical platform capabilities for future hardware.
- Defined product direction for on-device ML and LLM optimizations within strict thermal, latency, and power constraints.
- Drove product planning for cross-platform machine-perception capabilities, integrating novel sensor inputs into spatial computing experiences.

Apple

Senior Engineering Program Manager, AI/ML

Oct 2023 - Oct 2024

- Led product strategy for device-integrated Siri and AI/ML experiences across Home Devices and unreleased hardware platforms, focused on latency, reliability, and architectural readiness.
- Defined product and technical direction for Siri on unreleased hardware, unifying requirements across 30+ engineering teams for Day-1 readiness.
- Established a cross-functional operating model for 120+ stakeholders across Engineering, Design, Marketing, and leadership, improving dependency visibility and feature execution.
- Translated AI/ML, latency, privacy, and platform constraints into product requirements, roadmap tradeoffs, and launch decisions.

Apple

Senior Engineering Program Manager, Vision Products Group

Cupertino, CA
Oct 2021 - Oct 2023

- Led product strategy and execution for pre-launch Apple Vision Pro platform software capabilities, framework migration, software update/recovery flows, and service readiness.
- Defined roadmap, success metrics, and readiness mechanisms for low-level platform capabilities, improving visibility into adoption, activation, quality, and launch risk.
- Drove migration planning for key visionOS frameworks and APIs ahead of launch, aligning engineering owners, readiness criteria, and adoption milestones.
- Led product definition for visionOS software update and recovery mechanisms, partnering with Design and Engineering to reduce user friction and improve serviceability.
- Established operational processes for AppleCare and factory repair workflows, enabling service readiness for a brand-new product category.

Apple

Engineering Program Manager, Software Engineering

Nov 2019 - Oct 2021

- Led cross-functional prioritization for factory software supporting Apple Silicon Macs, Magic Keyboard, and AirPods from concept through launch.
- Drove readiness for core security infrastructure across iOS, macOS, watchOS, and visionOS manufacturing lines, helping scale reliable production workflows.
- Built automated reporting pipelines to identify, triage, and escalate software blockers across factory lines, reducing time to resolution for critical manufacturing issues.
- Partnered with software engineering, operations, security, and manufacturing teams to translate production constraints into roadmap priorities and launch readiness plans.

Intel

Product Manager, Data Protection

Chandler, AZ
Mar 2016 - Nov 2019

- Led product delivery for enterprise data protection solutions spanning data classification, DLP, and encryption workflows.
- Led a cross-functional team of 10+ developers and system engineers, driving 54% year-over-year growth in requests serviced by the encryption infrastructure.
- Released multiple versions of proprietary SDKs that expanded automation adoption across enterprise environments.

Apttus

Salesforce Analyst

San Mateo, CA
Nov 2015 - Mar 2016

- Provided premium support to high-value customers by debugging complex technical issues related to CPQ and CLM offerings.
- Developed troubleshooting guides and process articles to accelerate customer implementations.

Education

Santa Clara University

MS, Computer Science & Engineering

Santa Clara, CA

Illinois Institute of Technology

BS, Computer Science

Chicago, IL